

(Q1) (A) $V_{CC} : 15\text{ V}$ and $V_{EE} : -6\text{ V}$

(B) When gate signal is changing from V_{CC} to $-V_{EE}$

$$5 R_2 C = 1 \mu\text{s}$$

$$R_2 = \frac{1 \times 10^{-6}}{5 \times 10 \times 10^{-9}}$$

$$R_2 = 20 \Omega$$

When gate signal is changing from $-V_{EE}$ to V_{CC}

$$5 \times \frac{R_1 R_2}{R_1 + R_2} C = 0.3 \mu\text{s}$$

$$\frac{5 \times 20 \times R_1 \times 10 \times 10^{-9}}{R_1 + 20} = 0.3 \times 10^{-6}$$

$$R_1 = 8.57 \Omega$$

(C) Peak current through Q_1 , $I_{pk, Q_1} = \frac{V_{CC} - V_{EE}}{R_1 \parallel R_2}$

$$R_1 \parallel R_2 = 6 \Omega$$

$$\therefore I_{pk, Q_1} = \frac{15 - (-6)}{6}$$

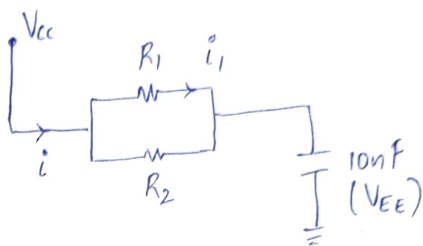
$$I_{pk, Q_1} = 3.5 \text{ A}$$

Peak current through Q_2 , $I_{pk, Q_2} = \frac{V_{CC} - V_{EE}}{R_2} = \frac{15 - (-6)}{20}$

$$I_{pk, Q_2} = 1.05 \text{ A}$$

(d) Yes, satisfactory because the settling times are much smaller than the time period (50 μ s).

(e) Diode turn ON,



Average power dissipation in R_1 :-

$$P_{R_1} = \frac{1}{T} \int i_1^2 R_1 dt$$

$$= \frac{1}{T} \int \left(i \cdot \frac{R_2}{R_1 + R_2} \right)^2 R_1 dt \quad \left[\because i_1 = i \cdot \frac{R_2}{R_1 + R_2} \right]$$

$$= \frac{1}{T} \int i^2 \cdot \frac{R_1 R_2}{R_1 + R_2} \cdot \frac{R_2}{R_1 + R_2} dt$$

$$= \left[\frac{1}{T} \int i^2 R_{eq} dt \right] \cdot \frac{R_2}{R_1 + R_2}$$

$$\text{Total average power } P_{T,avg} = \frac{1}{2} C (V_{CC} - V_{EE})^2 f_{sw}$$

$$\therefore P_{R_1} = \frac{R_2}{R_1 + R_2} \cdot P_{T,avg} = \frac{R_2}{R_1 + R_2} \cdot \frac{1}{2} C (V_{CC} - V_{EE})^2 f_{sw}$$

$$= \frac{20}{20 + 8.57} \cdot \frac{1}{2} \cdot 10 \times 10^{-9} (15 + 6)^2 \times 20,000$$

$$\boxed{P_{R_1} = 30.87 \text{ mW}}$$

$$\text{Similarly, } P_{R_2} = \frac{R_1}{R_1 + R_2} \cdot \frac{1}{2} C (V_{CC} - V_{EE})^2 f_{sw}$$

$$= \frac{8.57}{8.57 + 20} \cdot \frac{1}{2} \cdot 10 \times 10^{-9} \cdot (15 + 6)^2 \times 20,000$$

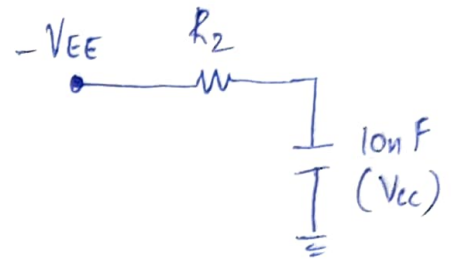
$$= 13.228 \text{ mW} \longrightarrow \textcircled{1}$$

During turning OFF,

$$P_{R_2} = \frac{1}{2} C (V_{EE} - V_{CC})^2 f_{sw}$$

$$= \frac{1}{2} \cdot 10 \times 10^{-9} \cdot (-6 - 15)^2 \times 20,000$$

$$= 44.1 \text{ mW} \longrightarrow \textcircled{2}$$

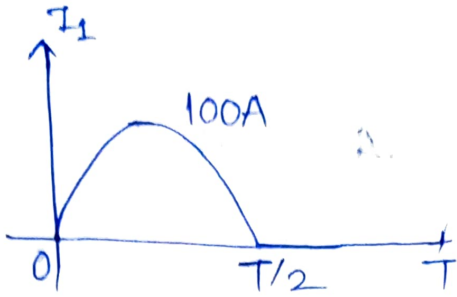


$\therefore$ Average power dissipation in R_2 :

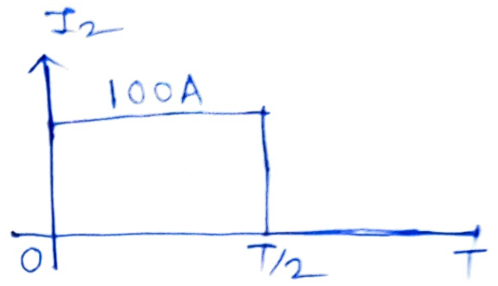
$$P_{R_2} = \textcircled{1} + \textcircled{2} = (13.228 + 44.1) \text{ mW}$$

$$P_{R_2} = 57.328 \text{ mW}$$

Q.2



①



②

Given,

For Waveform 1,

$$I_1 = 100 \sin \omega t = 100 \sin \frac{2\pi}{T} t$$

$$P_1 = 56.8 \text{ W}$$

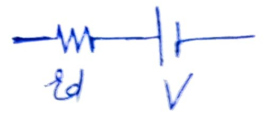
$$I_{1\text{avg}} = \frac{1}{T} \int_0^{T/2} 100 \sin\left(\frac{2\pi}{T} t\right) dt$$

$$= \frac{100}{T} \left[\frac{-\cos\left(\frac{2\pi}{T} t\right)}{\frac{2\pi}{T}} \right]_0^{T/2}$$

$$= \frac{100}{2\pi} \left[-\cos\left(\frac{2\pi}{T} \times \frac{T}{2}\right) + \cos 0 \right]$$

$$= \frac{100}{2\pi} \left[-(-1) + 1 \right]$$

$$I_{1\text{avg}} = \frac{100}{\pi} \text{ A}$$



SCR

$$\begin{aligned}
 I_{rms} &= \sqrt{\frac{1}{T} \int_0^{T/2} 100^2 \sin^2\left(\frac{2\pi t}{T}\right) dt} \\
 &= \sqrt{\frac{100^2}{T} \int_0^{T/2} \left(\frac{1 - \cos\frac{4\pi t}{T}}{2}\right) dt} \\
 &= \sqrt{\frac{100^2}{2T} \left(t - \frac{\sin\left(\frac{4\pi t}{T}\right)}{\frac{4\pi}{T}}\right) \Big|_0^{T/2}} \\
 &= \sqrt{\frac{100^2}{2T} \left[T/2 - \frac{\sin\left(\frac{4\pi \times T/2}{T}\right)}{\frac{4\pi}{T}}\right]} \\
 &= \sqrt{\frac{100^2}{2T} [T/2 - 0]} \\
 &= \sqrt{\frac{100^2}{4}} = \frac{100}{2} = 50
 \end{aligned}$$

$$I_{rms} = 50 \text{ A}$$

$$\text{Now, } P_1 = 56.8 \text{ W} = I_{rms}^2 R_d + I_{avg} \cdot V$$

$$\therefore 56.8 = 50^2 R_d + \frac{100}{\pi} V$$

$$56.8 = 2500 R_d + 31.83 \text{ V} \rightarrow \text{Eq (1)}$$

For waveform ②

$$I_{2avg} = \frac{1}{T} \int_0^{T/2} 100 \, dt = \frac{100}{T} (t)_0^{T/2}$$

$$I_{2avg} = \frac{100}{2} = 50 \, A$$

$$I_{2rms} = \sqrt{\frac{1}{T} \int_0^{T/2} 100^2 \, dt} = \sqrt{\frac{100^2}{T} (t)_0^{T/2}}$$
$$= \frac{100}{\sqrt{2}}$$

$$I_{2rms} = \frac{100}{\sqrt{2}} \, A$$

$$P_2 = I_{2rms}^2 \, R_d + I_{2avg} \cdot V$$

$$100 = \left(\frac{100}{\sqrt{2}}\right)^2 \cdot R_d + 50 \cdot V$$

$$100 = 5000 \cdot R_d + 50 \, V \quad \text{--- Eqn ②}$$

Solving eqn ① & eqn ②, we get,

$$R_d = 0.01 \, \Omega, \quad V = 0.9956 \, V$$

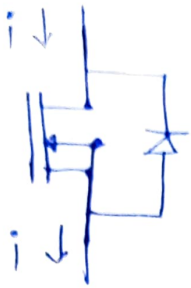
Ans:

A) $I_{1avg} = \frac{100}{\pi} \, A, \quad I_{1rms} = 50 \, A$

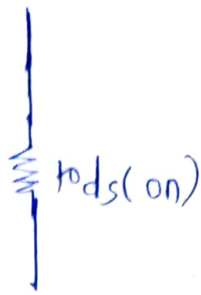
$I_{2avg} = 50 \, A, \quad I_{2rms} = \frac{100}{\sqrt{2}} \, A$

B) $R_d = 0.01 \, \Omega, \quad V = 0.9956 \, V$

9.3



Composite
Switch

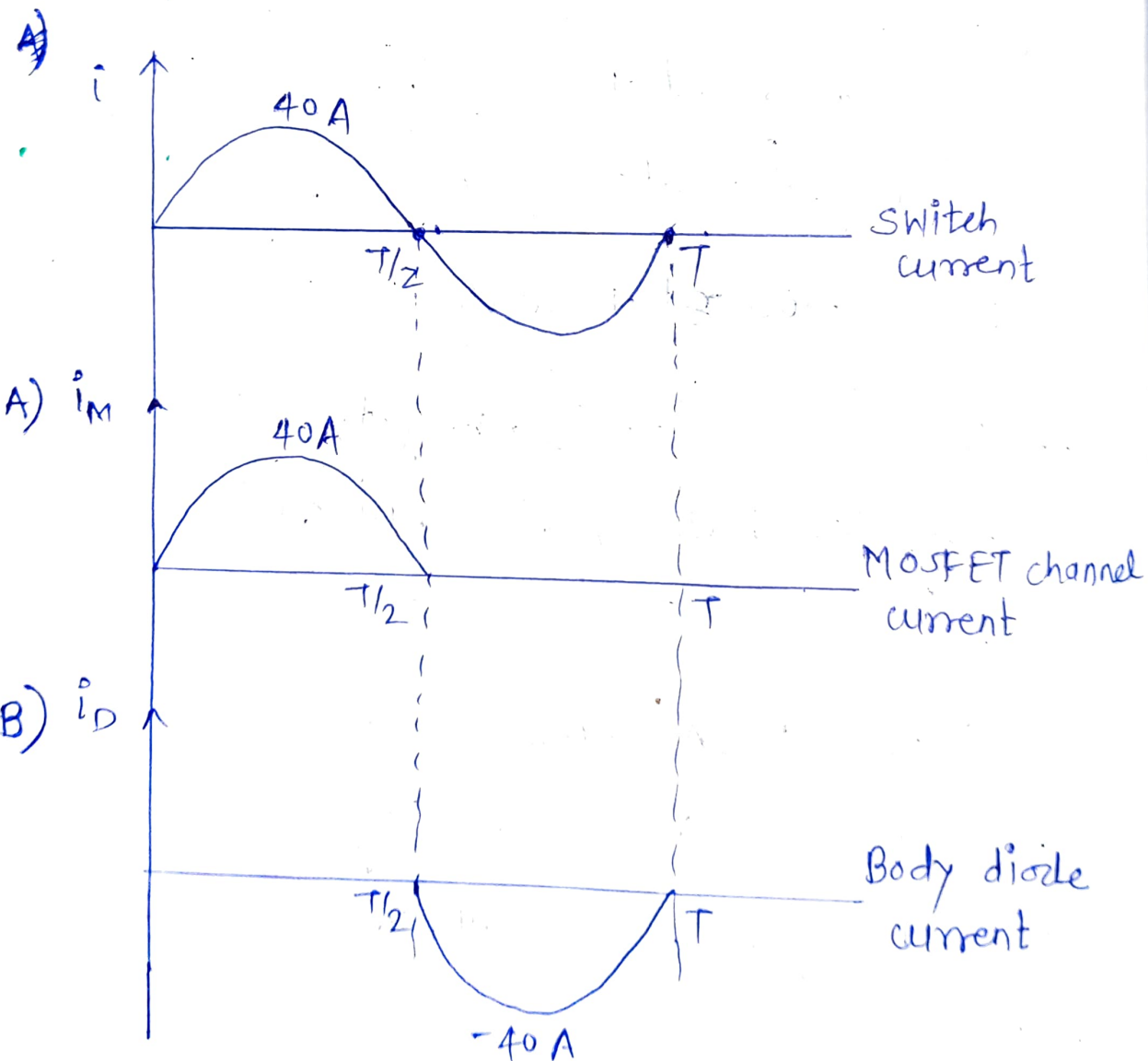


MOSFET model
in ON state



Body diode
model in
ON state

→ Positive current flows through MOSFET channel while negative current flows through body diode.



C) Average Power loss in MOSFET channel

$$P_M = I_{rms}^2 \times r_{ds(on)}$$

$$I_{rms} = \frac{40}{2} = 20 \text{ A}$$

$$P_M = 20^2 \times 50 \text{ m}\Omega = 400 \times 50 \text{ m}\Omega$$

$$P_M = 20 \text{ W}$$

D) Average Power loss in body diode

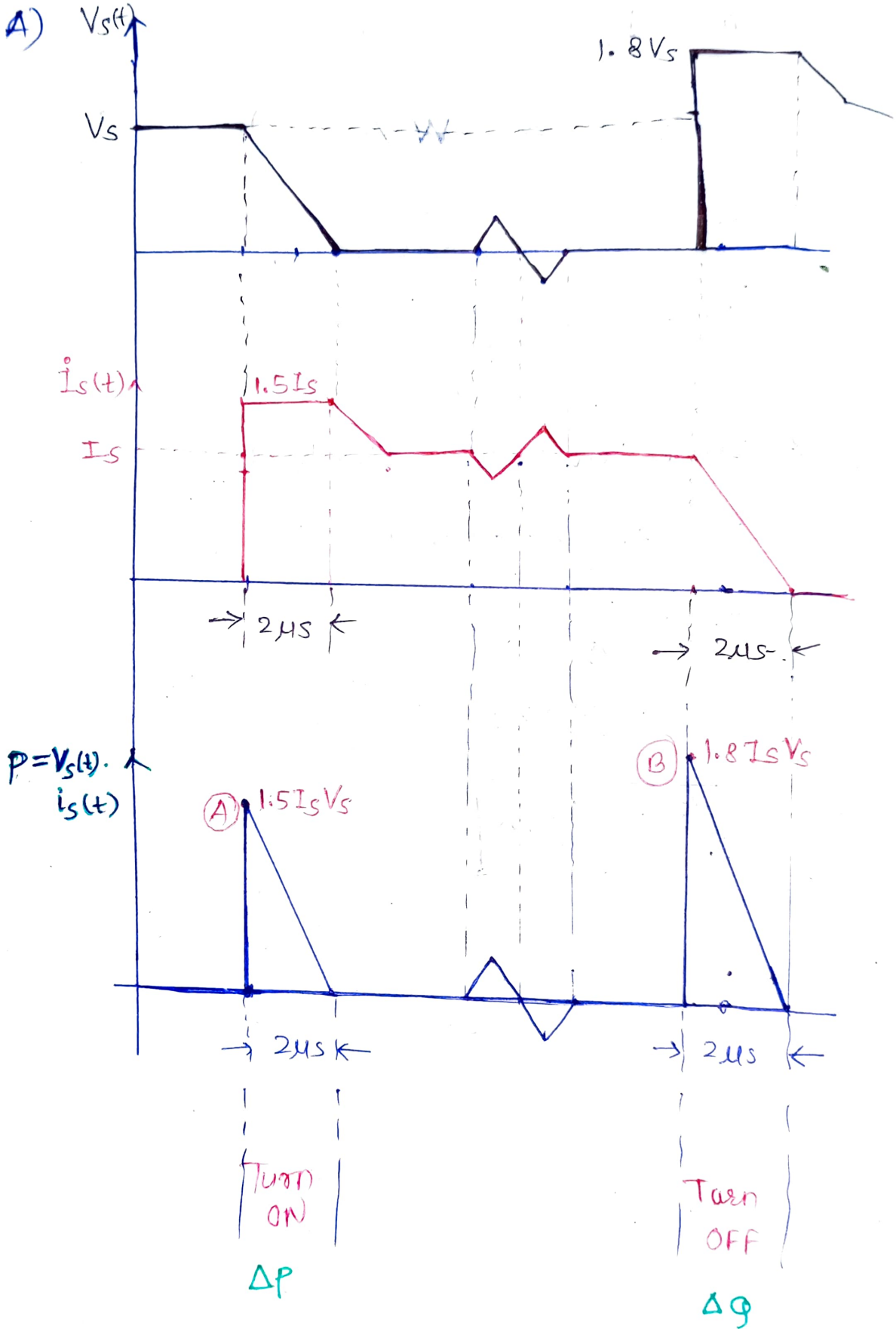
$$P_D = I_{Davg} \cdot V_{D(on)}$$

$$I_{Davg} = \frac{40}{\pi}$$

$$P_D = \frac{40}{\pi} \times 1$$

$$P_D = \frac{40}{\pi} \text{ W}$$

Q.4



B) Peak power during turn-on transition occurs at point A

$$\begin{aligned}\therefore A &= 1.5 I_S V_S = 1.5 \times 50 \text{ A} \times 200 \text{ V} \\ &= 15000 \text{ W} \\ &= 15 \text{ kW}\end{aligned}$$

a) Peak power during turn-off occurs at point B.

$$\begin{aligned}\therefore B &= 1.8 I_S V_S = 1.8 \times 10000 \\ &= 18 \text{ kW}\end{aligned}$$

$$\begin{aligned}\text{c) Turn-on energy loss} &= \text{Area of } \Delta P \\ (E_{ON}) &= \frac{1}{2} \times 2 \mu\text{s} \times A \\ &= \frac{1}{2} \times 2 \times 10^{-6} \times 15000 \\ &= 15 \text{ mJ}\end{aligned}$$

$$\begin{aligned}\text{Turn-off energy loss} &= \text{Area of } \Delta Q \\ (E_{OFF}) &= \frac{1}{2} \times 2 \mu\text{s} \times B \\ &= 18 \text{ mJ}\end{aligned}$$

$\therefore$ Average switching power loss

$$\begin{aligned}P_{loss} &= [(E_{ON}) + (E_{OFF})] \times \text{switching freq.} \\ &= (15 + 18) \text{ mJ} \times 10 \text{ kHz}\end{aligned}$$

$$\boxed{P_{loss} = 330 \text{ W}}$$